Flutter New Project Creation

Step-by-step guide using Android Studio on Windows

What this guide covers

• Create a new Flutter project in Android Studio

• Choose project name, location and platforms

• Understand the important project files and folders

• Run the app on an emulator or physical Android phone

• Edit main.dart and use Hot Reload

• Create a project from the terminal

• Fix common project creation errors

Official documentation: Flutter documentation

1. Before you start

• Install Android Studio and the Flutter SDK first.

• In Android Studio, install the Flutter plugin (and Dart plugin if prompted).

• Run flutter doctor in a terminal and fix any Android toolchain issues.

• Start an emulator or prepare a physical Android phone for testing.

2. Create a new project in Android Studio

• Open Android Studio.

• On the Welcome screen, select New Flutter Project. If a project is already open, use File → New → New Flutter Project (the exact menu can vary by version).

• Select Flutter from the project type list.

• For Flutter SDK path, browse to your Flutter SDK folder, for example C:\src\flutter. This is the SDK folder, not its bin subfolder.

• Click Next.

3. Enter project details

Field	What to enter
Project name	Use lowercase letters and underscores, e.g. employee_app
Project location	Choose a folder where you have write permission, e.g. C:\src\projects
Description	Optional short explanation of the app
Organization	Use your company/domain identifier in reverse-domain style if applicable, e.g. com.example
Platforms	Select Android and any other platforms you plan to support

• Click Create or Finish.

• Wait while Android Studio creates the files and initializes the project. The first Gradle sync may take time because dependencies need to download.

4. Important files and folders

File/folder	Purpose
lib/main.dart	Main Dart entry file; start learning and editing here
pubspec.yaml	Project metadata, dependencies, assets and fonts
android/	Android-specific project files and Gradle configuration
test/	Automated tests
build/	Generated build output; usually not edited manually

5. Run your Flutter app

• Start your Android emulator from Device Manager, or connect your phone with USB debugging enabled.

• Wait until the device is fully started.

• Select the device from the device dropdown in Android Studio.

• Click the green Run ■ button.

• The first build may take a few minutes. Check the Run or Debug console if an error appears.

6. Edit your first screen

Open lib/main.dart and replace its content with this small example:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        appBar: AppBar(title: const Text('My First App')),
        body: const Center(
          child: Text('Hello Flutter!'),
        ),
      ),
    );
  }
}
```

Save the file. If the app is already running, use Hot Reload (lightning icon) to update the UI without restarting the whole app.

7. Create a project using the terminal (alternative)

Open Command Prompt or the Android Studio terminal and run:

```
cd C:\src\projects
flutter create employee_app
cd employee_app
flutter pub get
flutter run
```

Use a valid project name with lowercase letters and underscores. If you create the project in the terminal, open its folder in Android Studio; choose the folder that contains pubspec.yaml.

8. Common problems

Problem	What to try
New Flutter Project not visible	Install/enable the Flutter plugin and restart Android Studio.
Flutter SDK path invalid	Select the root Flutter SDK folder, e.g. C:\src\flutter, not C:\src\flutter\bin.
Gradle sync/build fails	Check internet access, Android SDK setup and the error details in the Build console.
No device available	Start an emulator or connect a phone, then check flutter devices.
Project name error	Use lowercase letters and underscores, e.g. my_first_app.

9. Final checklist

■ Flutter SDK path is correct.

■ Project name and location are valid.

■ Project creation and Gradle sync finish.

■ Emulator or phone is detected.

■ The app runs successfully.

■ You can edit lib/main.dart and use Hot Reload.

Tip: Start by changing the AppBar title and Text widget, then practise Container, Row, Column, buttons and text fields.